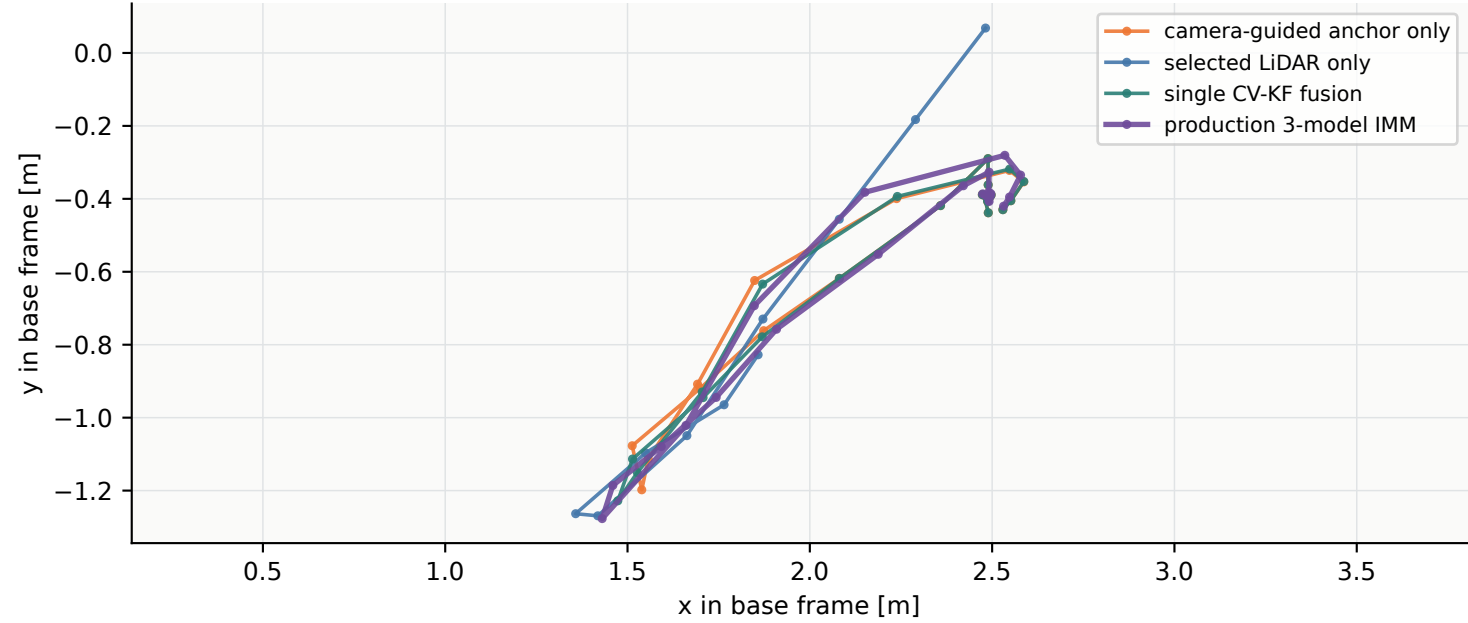
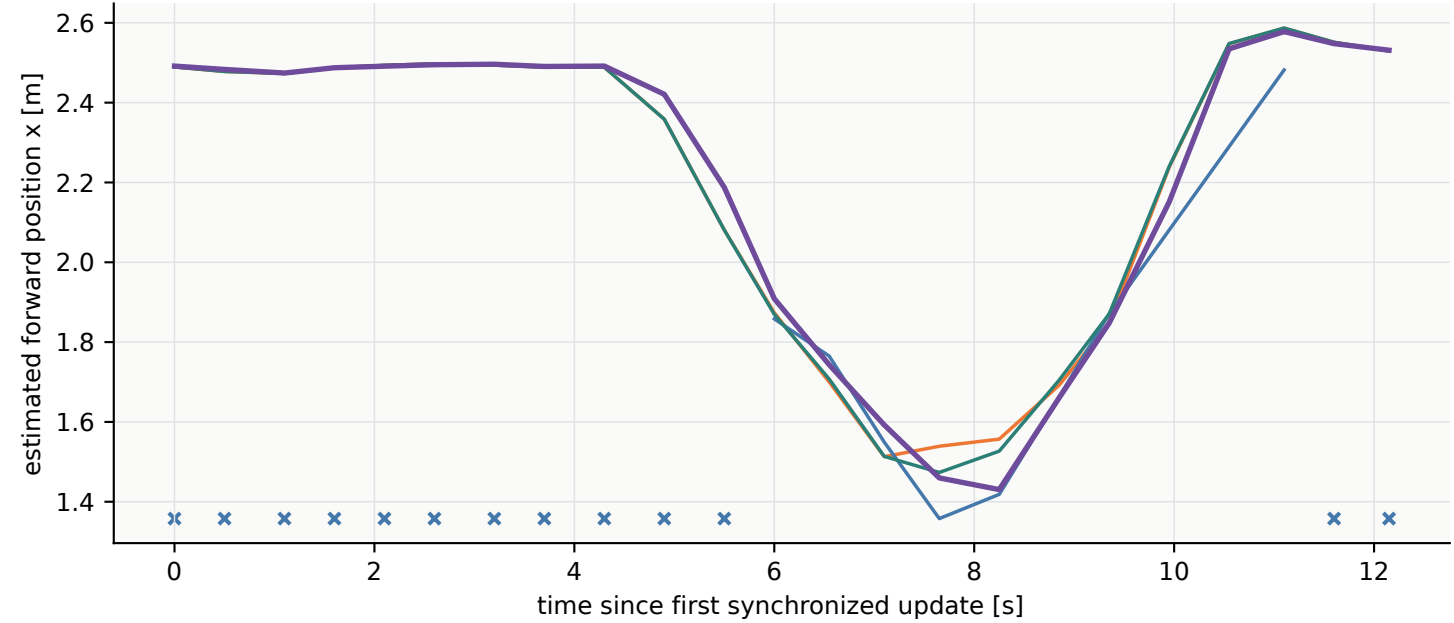


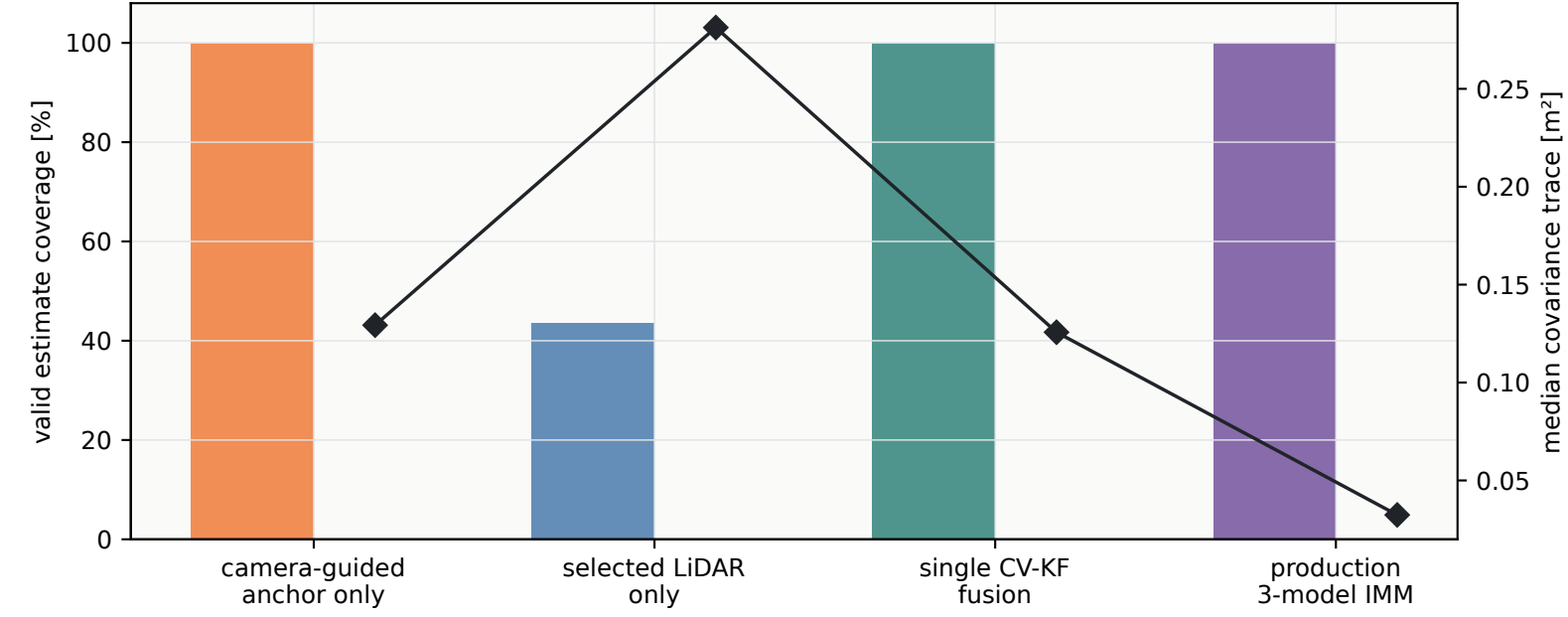
(a) Estimator trajectories under masked inputs



(b) Temporal continuity of the estimated target



(c) Continuity and estimator-reported uncertainty



(d) Consistency metrics and interpretation

condition	mutual dev. RMS [m]	accel. RMS [m/s²]
camera-guided anchor only	0.070	0.415
selected LiDAR only	0.168	0.510
single CV-KF fusion	0.062	0.402
production 3-model IMM	reference	0.461

All conditions reuse production target authorization and association. Camera-only uses a camera-guided LiDAR depth anchor; it is not monocular 3D. Mutual deviation is relative to the production IMM, not trajectory error.

DATA	TIMEOUT	REFERENCE	CLAIM
23 synchronized estimator updates from one replay	2.0 s measurement-age validity	production three-model IMM	estimator-input ablation; no external ground truth